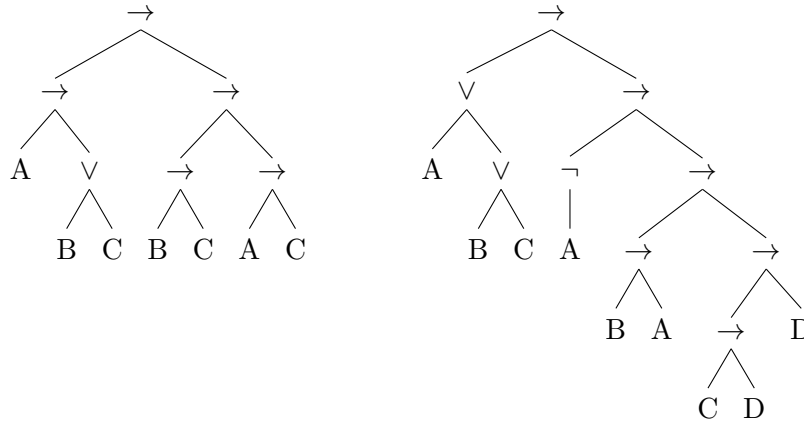


Exercise Sheet 12d - Solutions

Propositional Logic – Natural Deduction

1. The parse trees of F and G are



2. Here is a constructive Natural Deduction proof of $(A \rightarrow (B \vee C)) \rightarrow (B \rightarrow C) \rightarrow A \rightarrow C$

$$\begin{array}{c}
 \frac{\overline{A \rightarrow (B \vee C)}^1 \quad \overline{A}^3}{B \vee C} \quad [\rightarrow E] \quad \frac{\overline{B \rightarrow C}}{C}^2 \quad \frac{\overline{C}}{C \rightarrow C}^4 \quad [\vee E] \\
 \frac{\overline{C}}{A \rightarrow C}^3 [\rightarrow I] \\
 \frac{\overline{A \rightarrow C}}{(B \rightarrow C) \rightarrow A \rightarrow C}^2 [\rightarrow I] \\
 \frac{\overline{(B \rightarrow C) \rightarrow A \rightarrow C}}{(A \rightarrow (B \vee C)) \rightarrow (B \rightarrow C) \rightarrow A \rightarrow C}^1 [\rightarrow I]
 \end{array}$$

3. Here is a constructive Natural Deduction proof of $(A \vee B \vee C) \rightarrow \neg A \rightarrow (B \rightarrow A) \rightarrow (C \rightarrow D) \rightarrow D$

$$\begin{array}{c}
 \overline{A \vee B \vee C}^1 \quad \frac{\overline{A}^5 \quad \overline{\neg A}^2}{\perp}^2 [\neg E] \quad \frac{\perp}{D}^5 [\perp E] \quad \overline{B \vee C}^6 \quad \frac{\overline{B \rightarrow A}^3 \quad \overline{B}^7}{A}^7 [\rightarrow E] \quad \frac{A}{\neg A}^2 [\neg E] \\
 \frac{\perp}{D}^5 [\perp E] \quad \frac{\perp}{B \rightarrow D}^7 [\rightarrow I] \quad \frac{B \rightarrow D}{(B \vee C) \rightarrow D}^6 [\rightarrow I] \quad \frac{(B \vee C) \rightarrow D}{C \rightarrow D}^4 [\vee E] \\
 \frac{\overline{A \vee B \vee C}^1 \quad \overline{A \rightarrow D}^5}{D}^5 [\rightarrow I] \quad \frac{D}{(C \rightarrow D) \rightarrow D}^4 [\rightarrow I] \\
 \frac{\overline{(C \rightarrow D) \rightarrow D}}{(B \rightarrow A) \rightarrow (C \rightarrow D) \rightarrow D}^3 [\rightarrow I] \\
 \frac{\overline{(B \rightarrow A) \rightarrow (C \rightarrow D) \rightarrow D}}{\neg A \rightarrow (B \rightarrow A) \rightarrow (C \rightarrow D) \rightarrow D}^2 [\rightarrow I] \\
 \frac{\overline{\neg A \rightarrow (B \rightarrow A) \rightarrow (C \rightarrow D) \rightarrow D}}{(A \vee B \vee C) \rightarrow \neg A \rightarrow (B \rightarrow A) \rightarrow (C \rightarrow D) \rightarrow D}^1 [\rightarrow I]
 \end{array}$$

4. F and G are equivalent as they are both true and therefore are both equivalent to $\top$.

Another explanation is that because both formulas are provable, by consistency they are both valid, which means that they are both true for all valuations, i.e., that they are equivalent.